TABLE OF CONTENTS

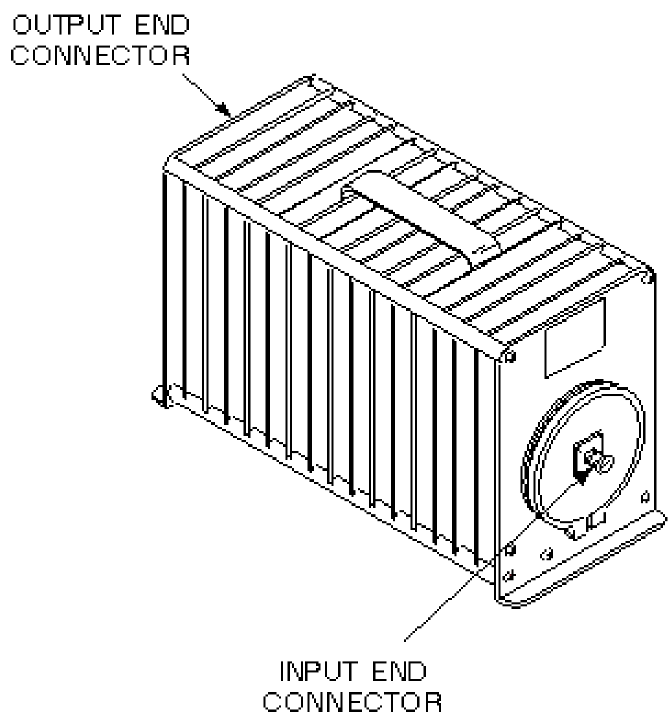
Description - There are two common failures to the Bird 30dB Dummy Load/Attenuator (46-258268P1-Product Dummy Load or 46-255837P10-Service Attenuator Tool), described as follows:

The most common failure of the 30 dB Dummy Load is the output resistor R4. This failure occurs whenever the RF amplifier is connected to the OUTPUT of the dummy load instead of the INPUT, thus destroying the output resistor R4. Replacement is simple and takes less than one hour.

Damage can occur to the large input resistor, R1. This can occur during large RF power duty cycles, or fatigue failure of the resistor. Arcing can cause the carbon in the resistor coating to disintegrate. Replacement is simple and can take less than one hour.

1- DETERMINE FAILURE CONDITION

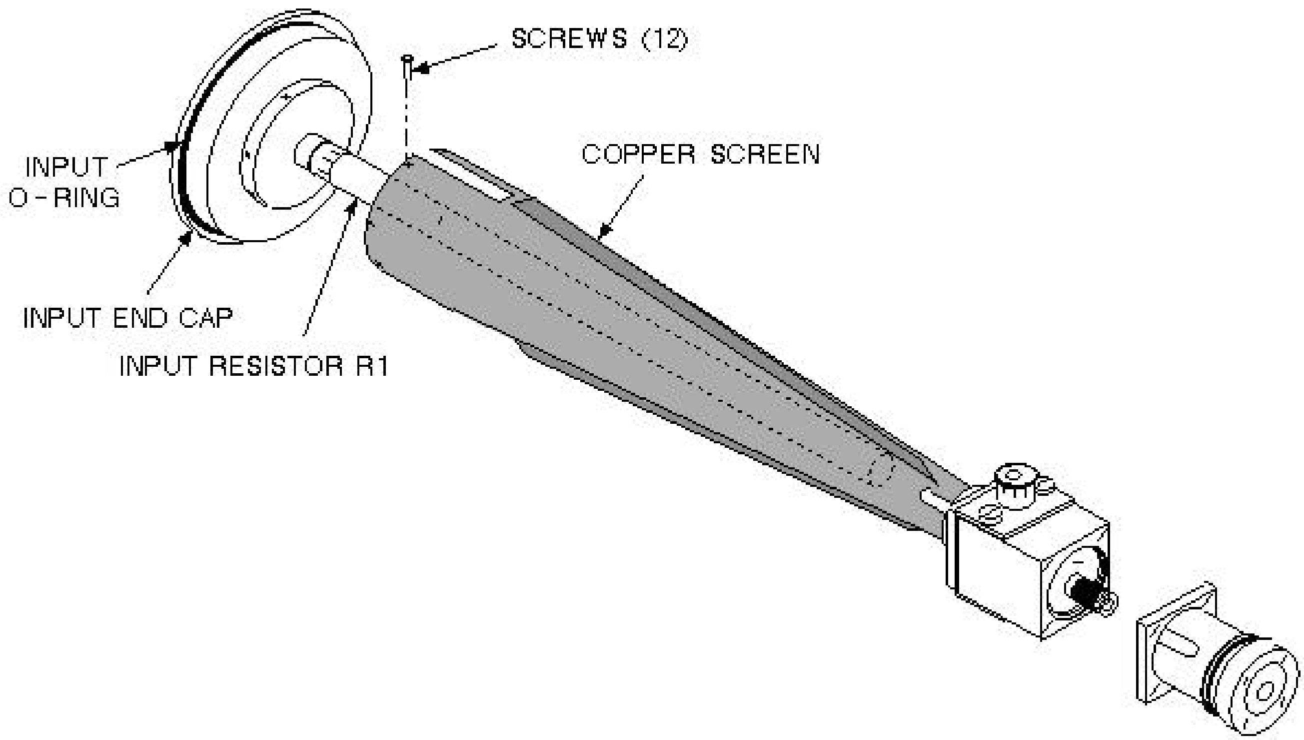
1. Measure the input resistance of the dummy load from the center pin of the input N connector to the outside of the connector. Verify that the resistance is approximately 50 ohms. See Illustration L2933A.



BIRD DUMMY LOAD/ATTENUATOR
ILLUSTRATION L2933A

2. If the input resistance reads open, then either R1, R2, or R3 are open (It is highly unlikely that R2 or R3 are open). Thus it is most likely that R1 (input resistor) needs to be replaced.

3. To test R2 and R3, each resistor must be removed from the resistor housing. Remove the two screws from the plate holding each of the resistors in place. Pull the resistor out of the housing. Measure the continuity across each resistor. Verify that resistance measures approximately 6.5 ohms. See Illustration L2934A.



RESISTOR HOUSING
ILLUSTRATION L2934A

4. If R1 (input resistance) reads shorted, then arcing may have occurred internal to the dummy load (this is highly unlikely to happen). This can usually be fixed by completely disassembling the dummy load, cleaning off any carbon tracks on the resistor and housing, then replacing the high dielectric oil. Verify that the input resistance is 50 ohms, as done in step 1.

5. Measure the output resistance of the dummy load from the center pin of the output N or BNC connector to the outside of the connector. Verify that resistance measures approximately 50 ohms. See Illustration L2933A above.

6. If the output resistance reads open, then R4 (output resistor) is open and must be replaced.

7. If R4 (output resistance) reads shorted, then resistor R4 may be damaged, or arcing may have occurred internal to the dummy load output section. This may be fixed by completely disassembling the dummy load, cleaning off any carbon tracks on the resistor and housing, then replacing the high dielectric oil. Verify that the input resistance is 50 ohms, as done in step 1.

2- TOOLS REQUIRED

TABLE 1
TOOLS AND REPLACEMENT PART REQUIRED

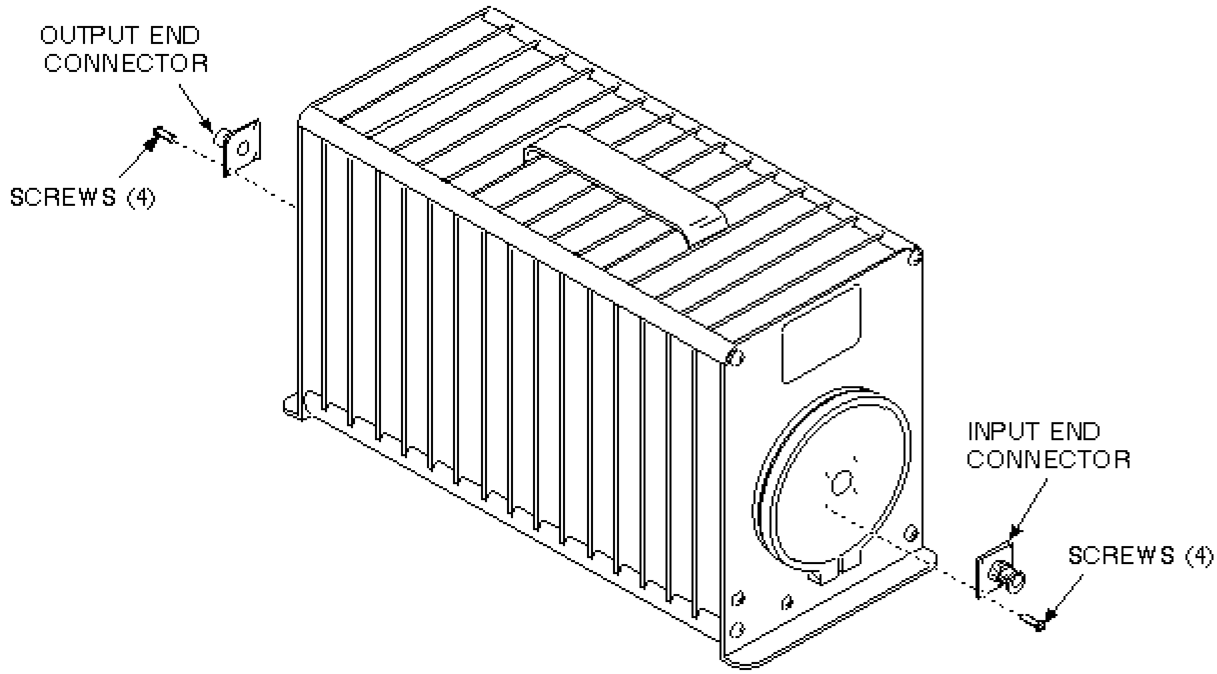
Item	Description	Part Number	Qty
1.	Transformer Oil (1 Gal. Can)	T0552G	1
2.	Dummy Load Repair Kit Note - Availability of Individual Parts	46- 317196G1	1
	Consisting of:		
	Output Resistor - 46.9 ohms (small resistor)	46- 317174P6	1
	Input Resistor - 52.6 ohms (large resistor)	46-317174P7	1
	Output O-Ring (small O-Ring)	46-317174P4	1
	Input O-Ring (large O-Ring)	46- 317174P3	1
	Spring Contact	46-317174P8	2
	Female BNC" Connector	46-317174P5	1
3.	Field Supplied Materials		
	Plastic Basin (3 Gal. or larger)	------	1
	Plastic Funnel	------	1
	Paper Towel (Box)	------	1

Note

Availability of individual part - Items in the Dummy Load Repair Kit (except for the female BNC connector) are available only in kit form, and cannot be ordered individually.

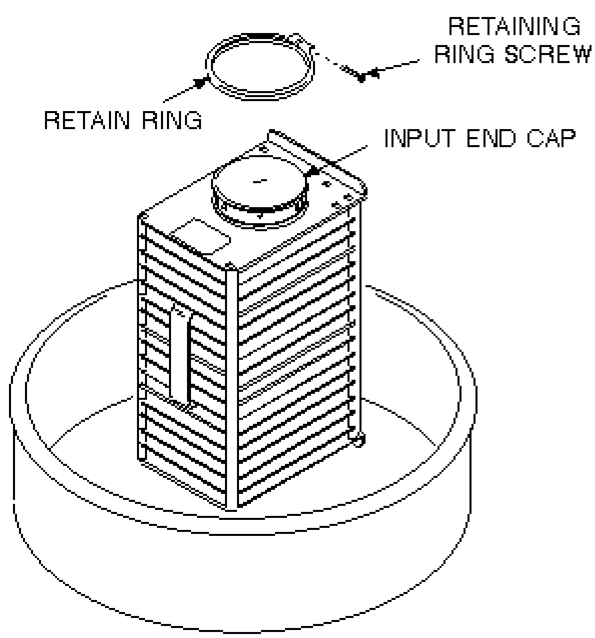
3- REMOVAL OF RESISTOR ASSEMBLY

1. Remove the four screws securing the two connectors (N and/or BNC) on the ends of the dummy load. Pull the connectors straight out from the dummy load. See Illustration L2935A.



REMOVAL OF END CONNECTORS
ILLUSTATION L2935A

2. Carefully stand the dummy load on the end marked OUTPUT, and position the dummy load in the plastic basin. See Illustration L2936A.



REMOVAL OF RESISTOR ASSEMBLY
ILLUSTRATION L2936A

3. Remove the retaining ring screw and retaining ring on the end marked INPUT of the dummy load. See Illustration L2936A above.

4. Exercising caution so as not to damage the rubber o-ring, carefully remove the end cap 1/4-inch. The end cap is usually very difficult to remove; rotating it during removal will help.

5. Pull the end cap out until the o-ring is clear of its sleeve, and the resistor assembly is free. Remove the resistor assembly from the dummy load housing and lay it on paper towel.

Note

Remove End Cap - As the end cap is removed, oil will run out the OUTPUT opening.

6. Use paper towel to wipe up any excess oil on the resistor assembly.

7. Check the color of the oil left in the plastic basin. If it has a yellow tint, it is bad. Replace it when you reassemble the dummy load after replacing the resistor(s.

Note

Dispose Of Old Dummy Load Oil - Dispose of the old dummy load oil as prescribed in GEMS Policy and Procedures, M207.000. Ensure that the oil is disposed of according to state and local ordinances.

8. With paper towel, wipe excess oil from the inside of the dummy load housing, and any carbon particles left in the oil from the resistors. Set the basin and dummy load housing aside.

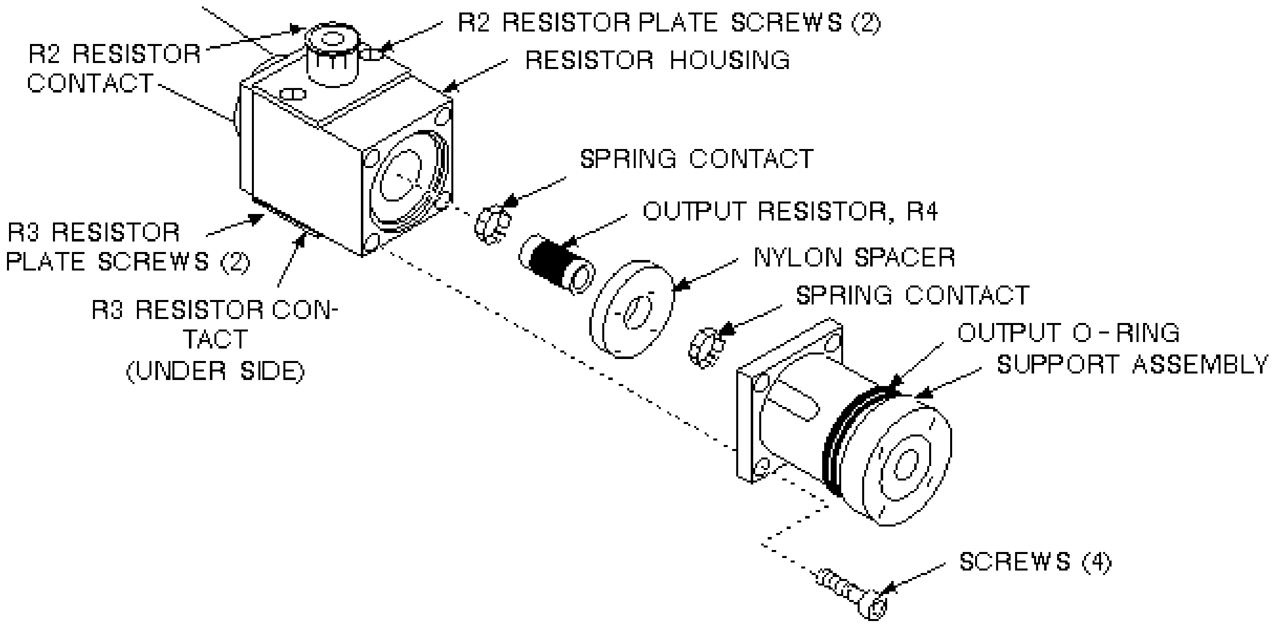
CAUTION

Possible equipment damage. The dummy load is filled with oil. Oil may drip out the Input or Output opening if the o-rings are damaged. Always replace both o-rings during repair procedures.

9. Inspect the output and input o-rings for cracks and loss of elasticity. Always replace both o-rings during repair procedures.

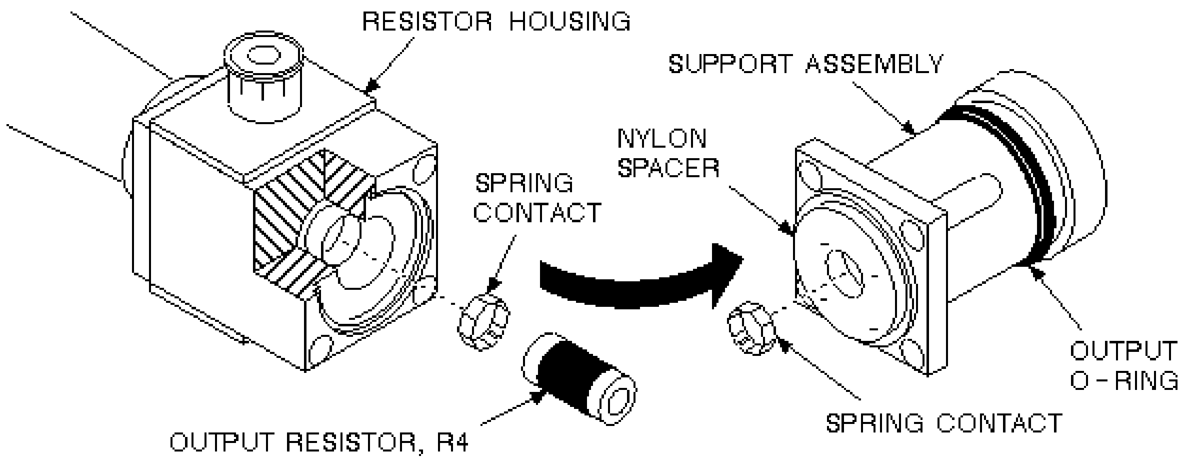
4- REPLACEMENT OF R4 (OUTPUT RESISTOR)

1. Remove the four allen screws at the OUTPUT end of the assembly, and remove the support assembly. See Illustration L2937A.



R4 (OUTPUT RESISTOR) REPLACEMENT
ILLUSTRATION L2937A

2. Remove the white nylon spacer and R4 (Output Resistor) and output o-ring. Discard R4 resistor and output o-ring. Remove the two spring contacts from R4 and/or support assembly. Replace the spring contacts (46-317174P8) with new ones supplied and discard the old spring contacts. See Illustration L2938A.



SPRING CONTACT REPLACEMENT
ILLUSTRATION L2938A

Note

R4 Resistor Inspection - Careful inspection of R4 resistor may reveal a slight crack around its surface which is an indication of failure.

3. Carefully install the new R4 resistor (46-317174P6) in the spring contact in the resistor housing. Ensure that the spring contact is not bent when you slip R4 into it.

Note

Spring Contacts Purpose - The spring contacts maintain proper connection between the R4 resistor and the connectors (i.e., the support assembly and resistor block).

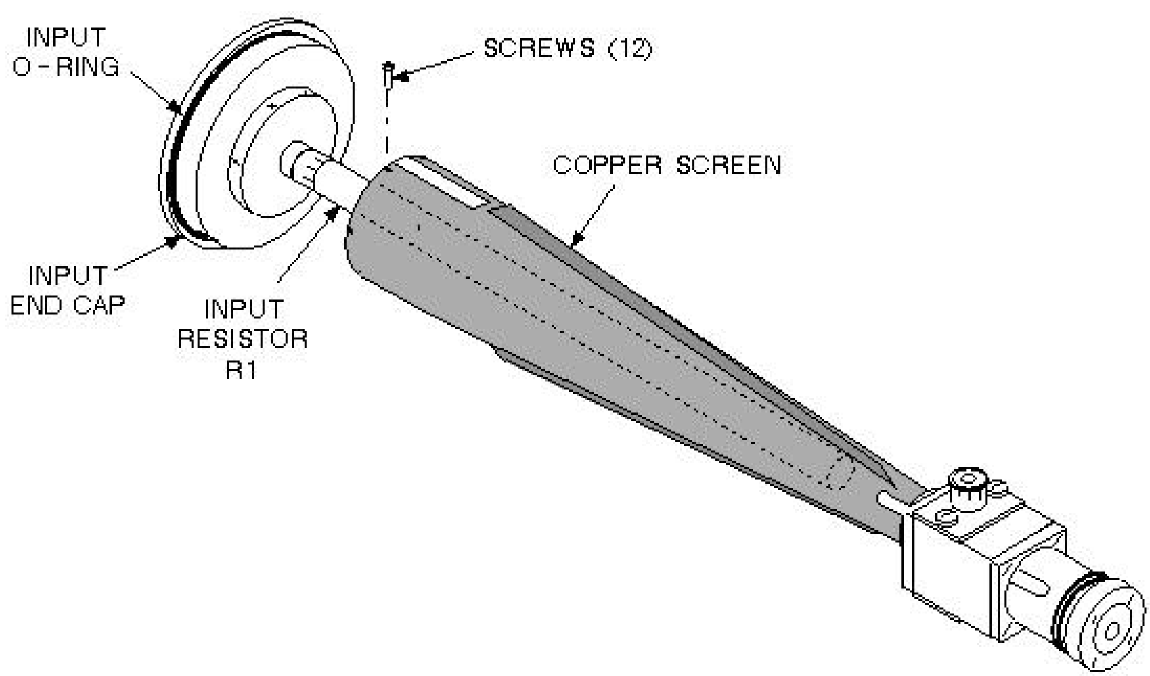
4. Install the new output o-ring (46-317174P4) onto the support assembly.

5. Carefully install the support assembly with the spring contact in place around the other end of R4 resistor.

6. Install the four allen screws to secure the support assembly to the resistor block.

5- REPLACEMENT OF R1 (INPUT RESISTOR)

1. Remove the twelve screws at the INPUT end of the assembly. See Illustration L2939A.



R1 (INPUT RESISTOR) REPLACEMENT
ILLUSTRATION L2939A

2. Carefully remove the end cap and R1 (Input Resistor) assembly from inside the copper screen. See Illustration L2939A above.

3. Remove R1 resistor and the input o-ring from the end cap. See Illustration L2939A above.

4. Clean away any excess oil or carbon particles from inside the connector wells in the under side of the end cap and in the bottom of the resistor assembly where the R1 resistor is inserted.

5. Install the new R1 resistor (46-317174P7) and the input o-ring (46-317174P3) on the end cap. See Illustration L2939A above.

6. Insert the end cap/R1 assembly into the copper screen and inside the connector well.

7. Reinstall the twelve screws holding the copper screen to the end cap.

6- REASSEMBLY OF RESISTOR ASSEMBLY

1. Carefully insert the resistor assembly, OUTPUT end first, into the dummy load housing until a 1/4-inch opening is left.

CAUTION

Possible equipment damage. Do not damage the o-ring. A thin coating of new transformer oil (T0552G) smeared on the o-ring will help slip the o-ring in place.

2. Place a funnel into the 1/4" opening. Reuse the high dielectric oil if the oil is not discolored YELLOW, or replace the discolored yellow oil with transformer oil (T0552G). Fill the dummy load housing to 1/2" from the top of the housing.

3. Install the resistor assembly flush to the dummy load housing.

Note

Tilt Dummy Load Housing - Dummy load housing must be tilted slightly to secure the INPUT end cap flush to the housing. This is because the OUTPUT end of the resistor assembly normally sticks out from the OUTPUT end cap.

4. Replace the retaining ring and screw on the INPUT end of the dummy load.

5. Replace the N connector on the INPUT end of the dummy load with the four screws.

6. Install a BNC connector (46-317174P5) on the OUTPUT end of the dummy load with the four screws. One is provided in the repair kit; the original N OUTPUT connector can be discarded.

7. Wipe away any excess oil from around the end caps, and check for any oil leaks.

7- FUNCTIONAL CHECKS REQUIRED

1. For dummy load calibration, refer to procedure for Dummy Load & Cables Calibration.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 21, 1998	J. Saperstein	Initial conversion from Toolbook to Word.